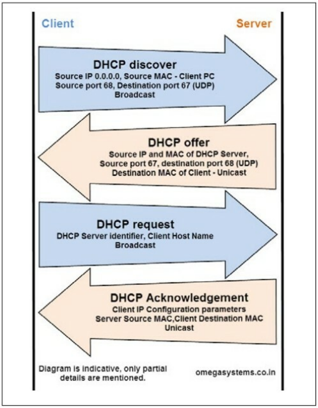

Figure 5: Screenshot of DHCP protocol

When a system configured with the 'Obtain an IP address automatically' setting is connected to a network, it uses DHCP to get an IP address from the DHCP server. Thus, this is a client–server protocol. To capture DHCP packets, users may start Wireshark on such a system, then start packet capture and, finally, connect the network cable.

Please refer to Figures 4 and 5, which give a diagram and a screenshot of the DHCP protocol, respectively.

Discovering DHCP servers: To discover DHCP server(s) in the network, the client sends a broadcast on 255.255.255.255 with the source IP as 0.0.0.0, using UDP port 68 (bootpc) as the source port and UDP 67 (bootps) as the destination. This message also contains the source MAC address as that of the client and ff:ff:ff:ff:ff:ff as the destination MAC.

A DHCP offer: The nearest DHCP server receives this 'discover' broadcast and replies with an offer containing the offered IP address, the subnet mask, the duration of the default gateway lease and the IP address of the DHCP server. The source MAC address is that of the DHCP server and the destination MAC address is that of the requesting client. Here, the UDP source and destination ports are reversed.

DHCP requests: Remember that there can be more than one DHCP server in a network. Thus, a client can receive multiple DHCP offers. The DHCP request packet is broadcast by the client with parameters similar to discovering a DHCP server, with two major differences:

1. The DHCP Server Identifier field, which specifies the IP address of the accepted server.
2. The host name of the client computer.

Use Pane 2 of Wireshark to view these parameters under 'Bootstrap Protocol' – Options 54 and 12.

The DHCP request packet also contains additional client requests for the server to provide more configuration parameters such as the default gateway, DNS (Domain Name Server), address, etc.

DHCP acknowledgement: The server acknowledges a DHCP request by sending information on the lease duration and other configurations, as requested by the client during the DHCP request phase, thus completing the DHCP cycle.

For better understanding, capture a few packets, use Wireshark 'Display Filters' to filter and view ARP and DHCP, and read them using Wireshark panes.

Saving packets

Packets captured using Wireshark can be saved from the menu 'File – Save as' in different formats such as Wireshark, Novell LANalyzer and Sun Snoop, to name a few.

In addition to saving all captured packets in various file formats, the 'File – Export Specified Packets' option offers users the choice of saving 'Display Filtered' packets or a range of packets.

Please feel free to download the pcap files used for preparing this article from *opensourceforu.com*. I believe all *OSFY* readers will enjoy this interesting world of Wireshark, packet capturing and various protocols!

Troubleshooting tips

Capturing ARP traffic could reveal ARP poisoning (or ARP spoofing) in the network. This will be discussed in more detail at a later stage. Similarly, studying the capture of DHCP protocol may lead to the discovery of an unintentional or a rogue DHCP server within the network.

A word of caution

Packets captured using the test scenarios described in this series of articles are capable of revealing sensitive information such as login names and passwords. Some scenarios, such as using ARP spoofing may disrupt the network temporarily. Make sure to use these techniques only in a test environment. If at all you wish to use them in a live environment, do not forget to get the explicit written permission before doing so. END

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